# Lab Report: Text classification with neural networks

## 1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 32. The learning rate was set to 0.0003 with the Adam optimizer.

## 2 Results

Figure 1 shows the training loss.

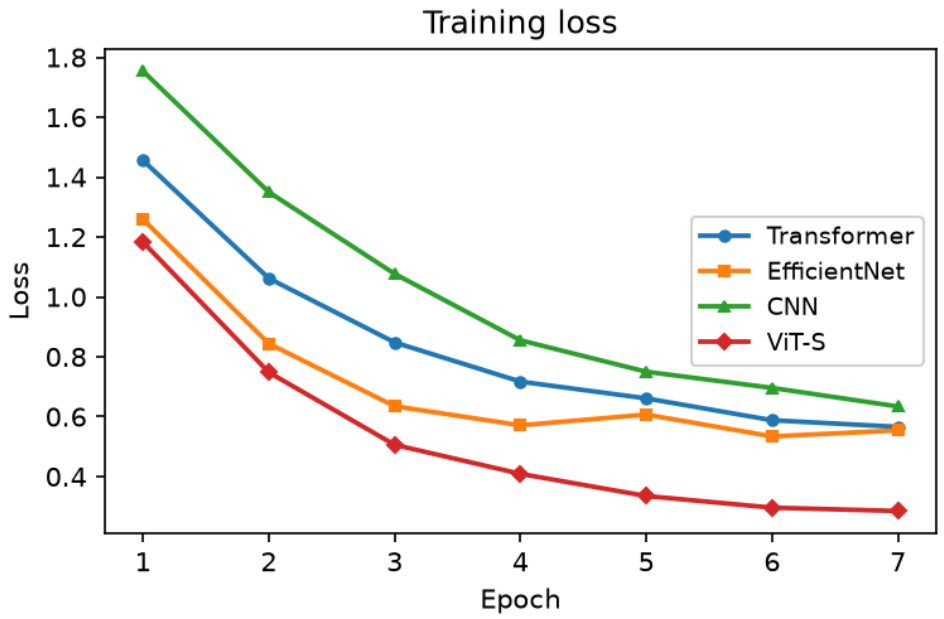



Figure 1: Training loss for the models in this experiment.

Transformer has a loss of about 1.5 at epoch 1. CNN has a loss of 1.754 at epoch 1.

Figure 2 shows the test accuracy during training.

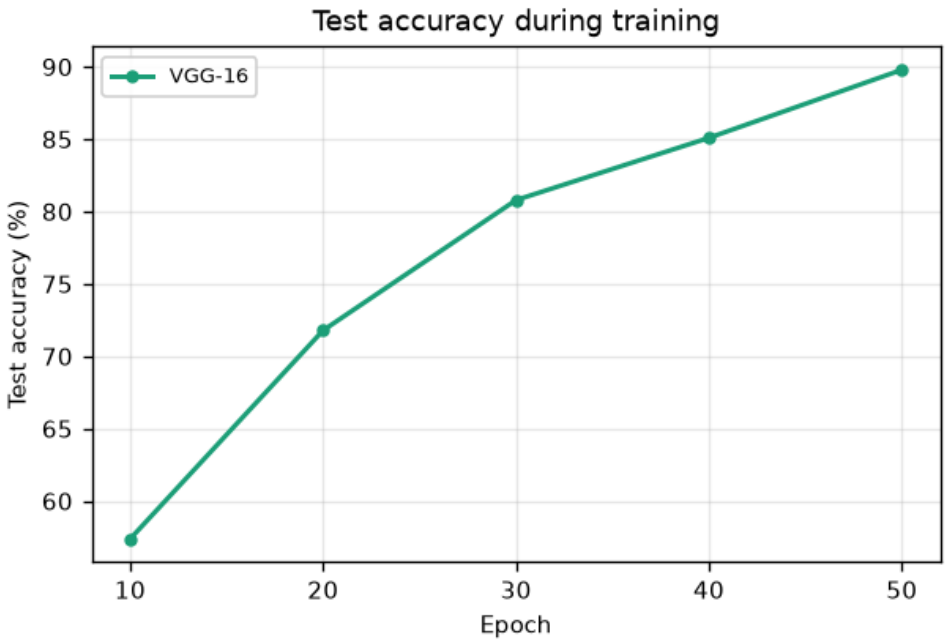



Figure 2: Test accuracy during training for the models in this experiment.

VGG-16 reaches over 75% test accuracy at epoch 30. VGG-16 reaches 57.4% test accuracy at epoch 10. VGG-16 scores about 9 points higher on 30 than on 20. VGG-16 attains a test accuracy of over 85% at

epoch 50.

Figure 3 shows the f1 score by dataset.

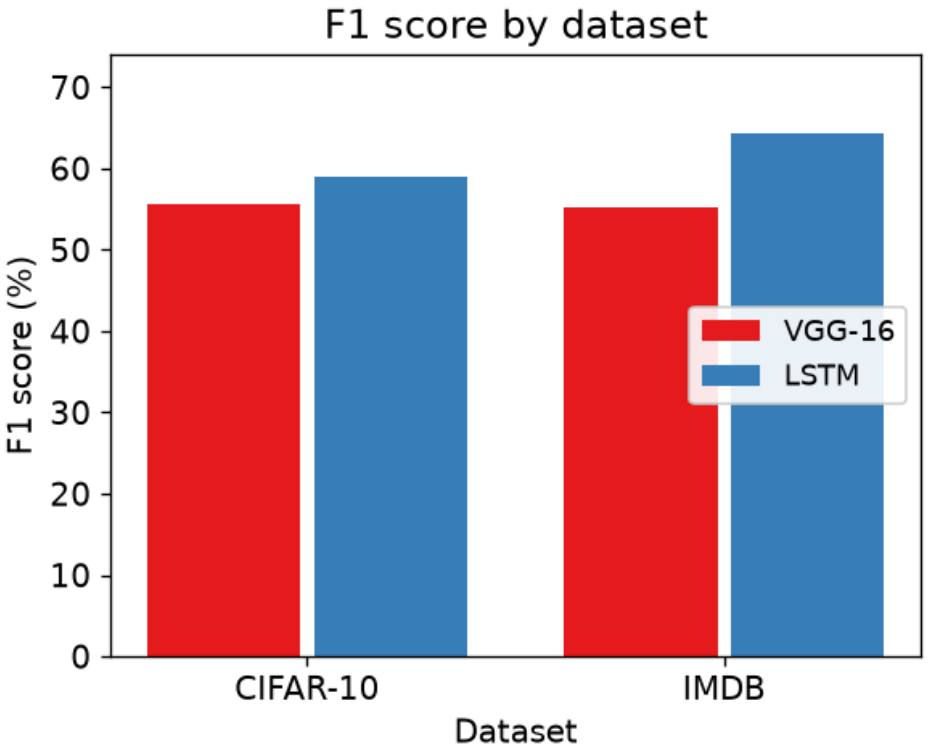



Figure 3: F1 score by dataset for the models in this experiment.

VGG-16 reaches 55.2% F1 score on IMDB. LSTM outperforms VGG-16 by 24.0 percentage points on IMDB (Figure 3). LSTM outperforms VGG-16 by about 9 percentage points on IMDB. On IMDB, LSTM obtains a F1 score of 101.4%.

## 3 Discussion

On QQP, SVM obtains an accuracy of 77.7%. Overall, the results are in line with what we expected from the literature.